

A Preliminary Approach to Investigate Periodicity in United States Tornado Activity: A Wavelet Analysis and Forecasting Approach using R



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Abstract

Historical records show a recent increase in tornado events within New York State, raising questions about a pattern tied to this trend. After previous research suggested the existence of a recurring 13–17 year cycle in tornado activity alongside a long-term upward trend in both minima and maxima, we sought to determine whether our findings remained consistent for the United States. Wavelet analysis detects frequency variations over time. After applying the Kolmogorov-Zurbenko (KZ) filter, which excludes high-frequencies, we were able to reveal annual and multi-annual cycles. We modelled our data using the Autoregressive Integrated Moving Average (ARIMA) function, allowing the general forecast of future tornado frequencies.

Introduction

Scientists have noticed a recent increase in tornado events in New York State, raising questions about the existence of a cyclical or trend-driven phenomenon driving these events. Beyond direct, immediate impacts, tornadoes are known to affect multiple geological processes, including soil erosion, sediment redistribution, and mass wasting.

Our previous research has suggested the existence of a 13–17 year cycle of tornado events in New York state. The confirmation of such a cycle nationwide could significantly advance our relatively limited understanding of tornadoes, aiding with preparational and risk mitigation related efforts.

Methods

- Data was collected from NOAA's Storm Events Database, which allows the bulk download of historical storm data across all fifty states. It was then converted to a binary system of 1's and 0's, denoting when a tornado did and didn't occur, respectively.
- A 30-day cumulative sum, the sum taken during any 30-day long interval, was taken and smoothed out with the KZ filter. This averaged the number of tornado occurrences over a given year, eliminating seasonal fluctuations.
- The wavelet function was used to identify dominant periodicities.
- Finally, the forecasting ARIMA model was used to generate the predicted US tornado frequencies over the next 2 years.

We further filtered and smoothed our data by increasing the range of our moving average to 7 years to account for the impacts of El Niño and La Niña on atmospheric conditions, which operate on an average range of 2 to 7 years.

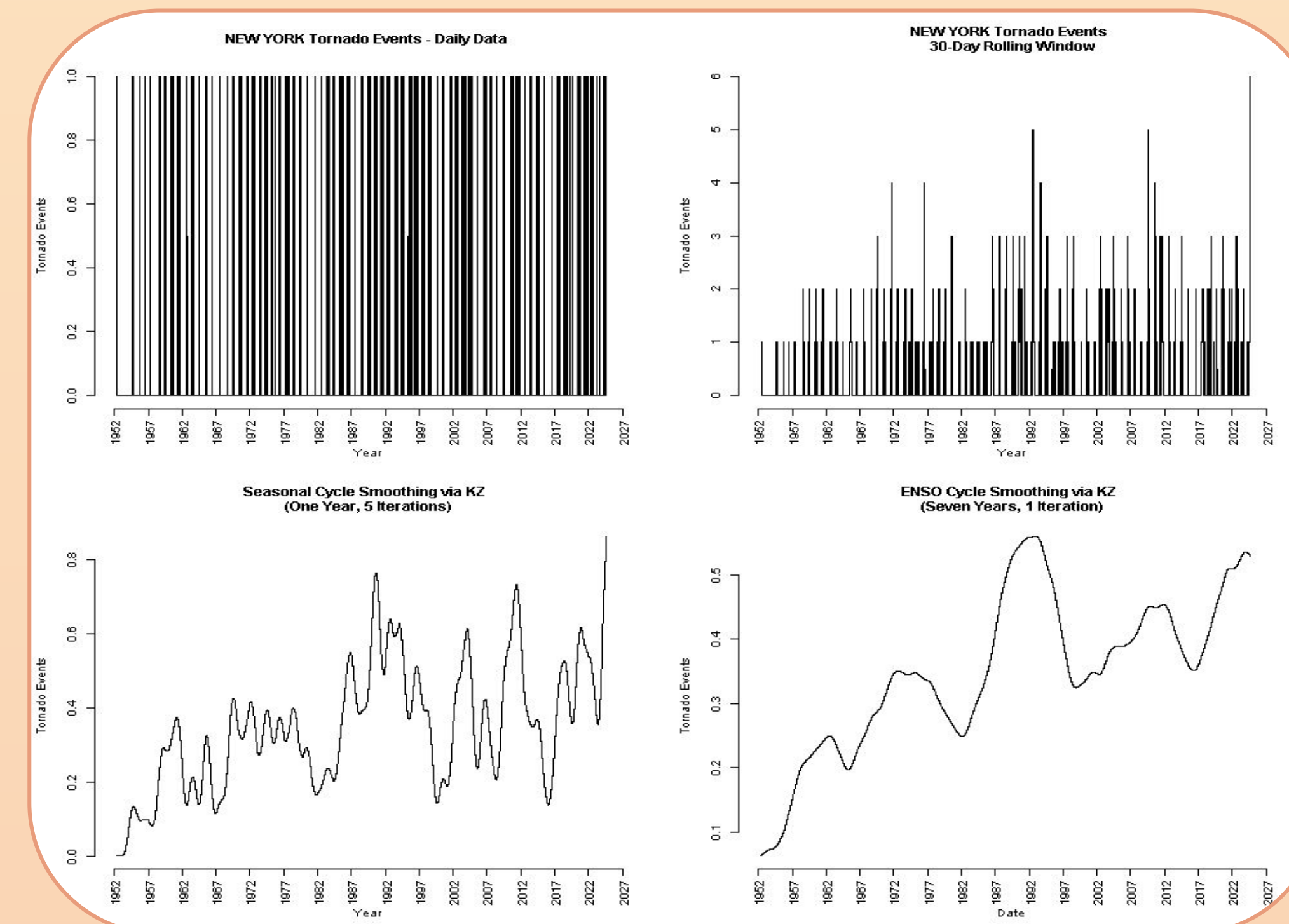


Figure 1: Demonstration of the stages of data processing for NYS Data.

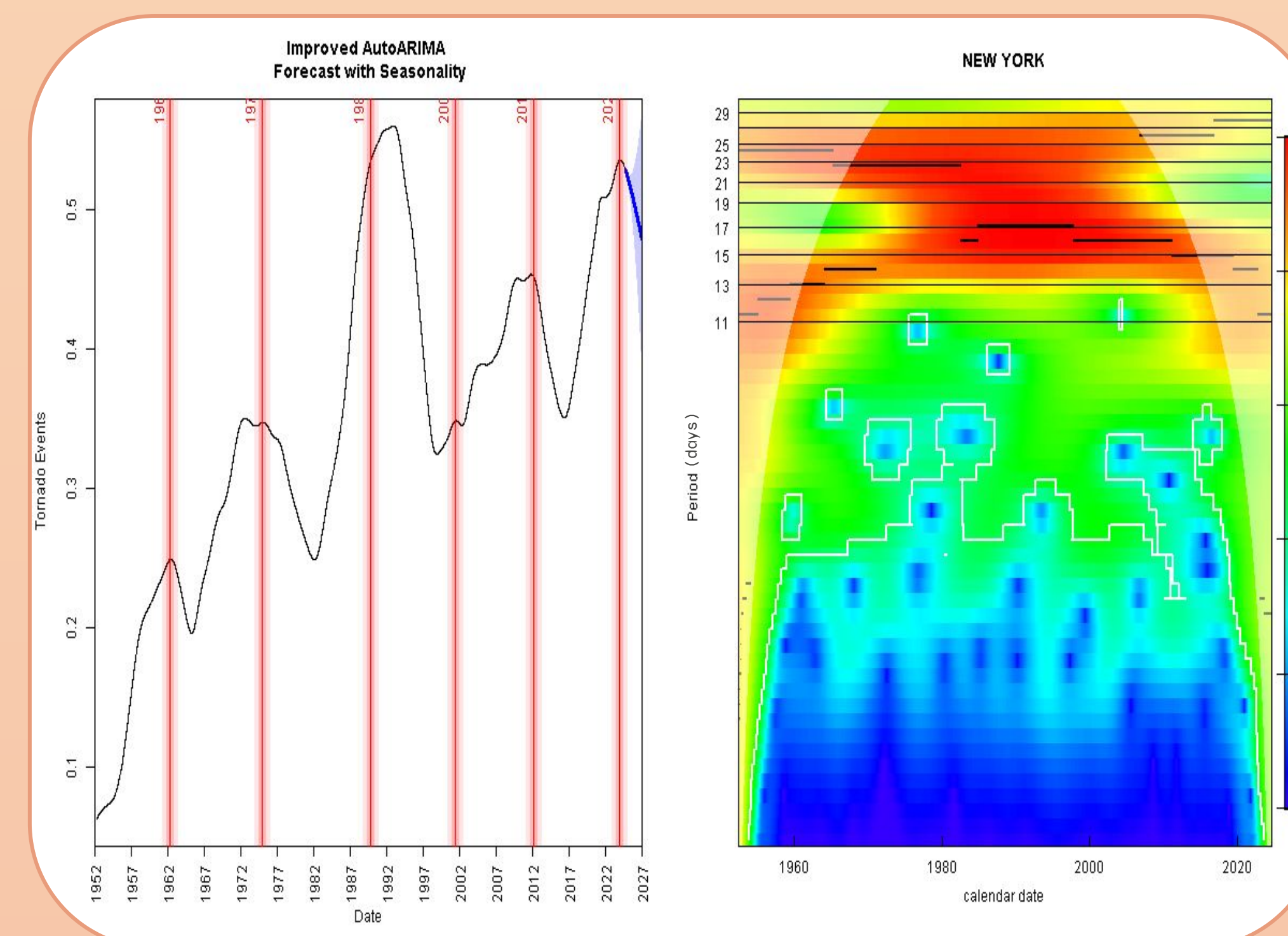


Figure 2: Demonstration of the stages of data processing for NYS Data.

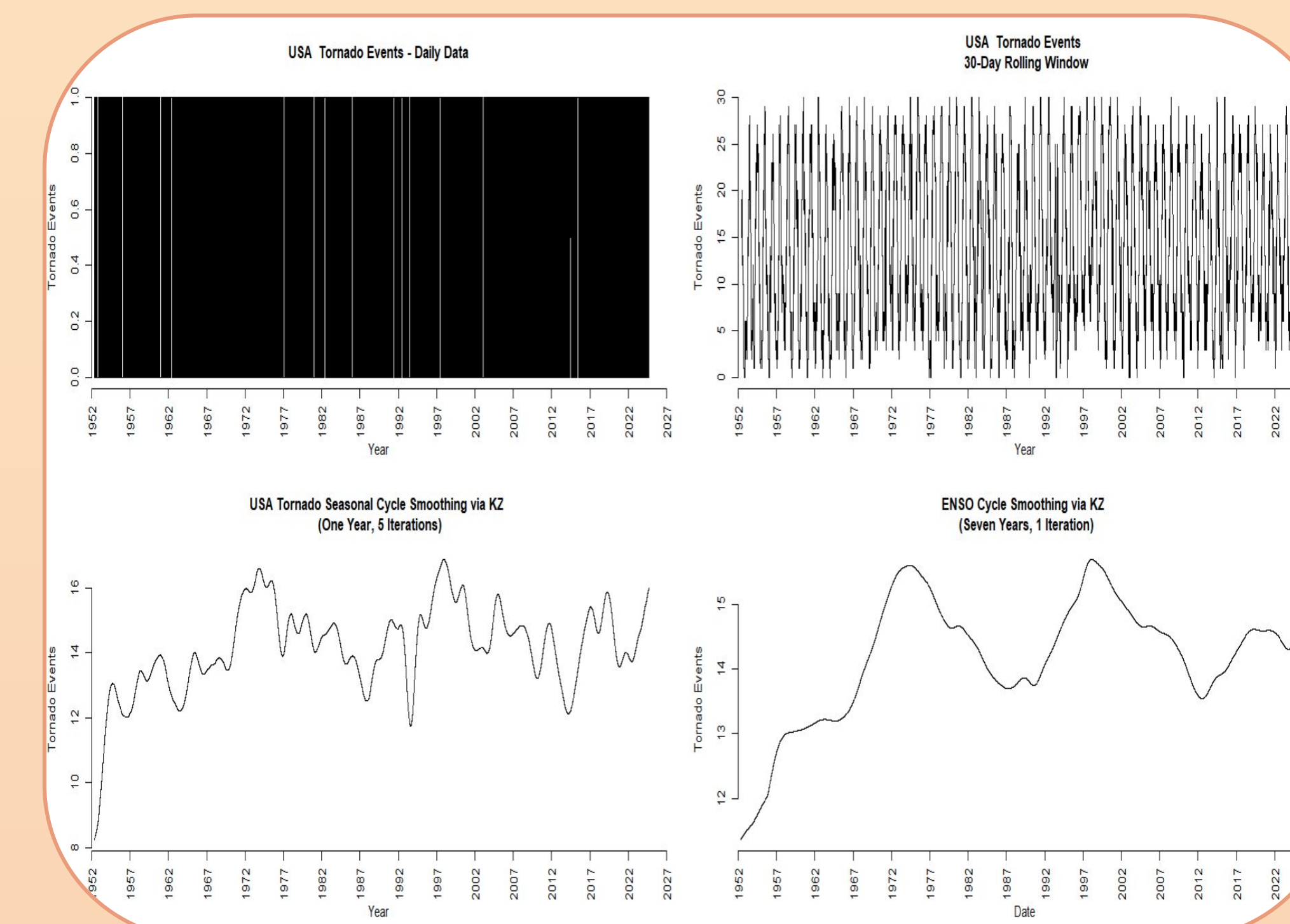


Figure 3: Demonstration of the stages of data processing for USA Data.

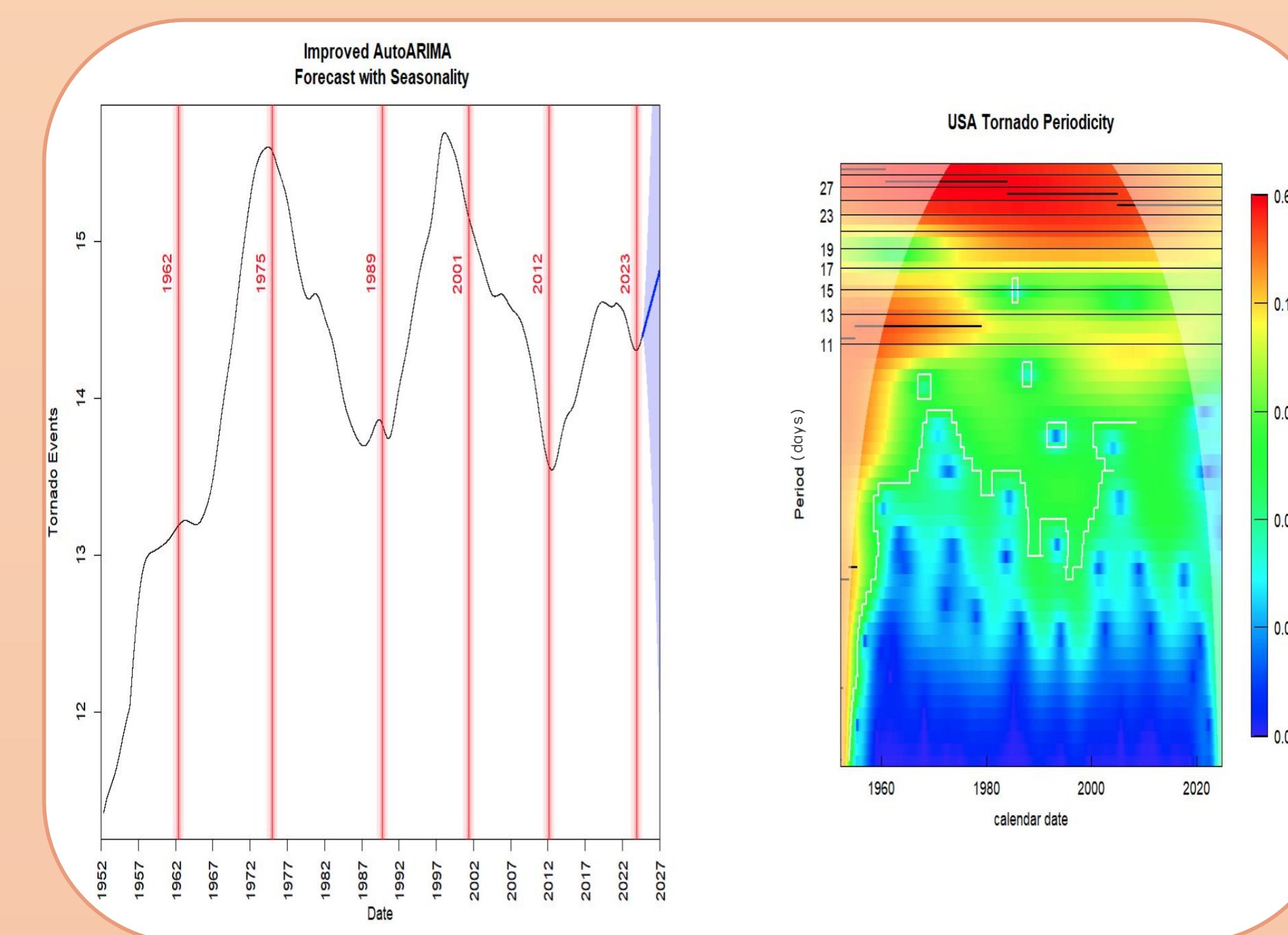


Figure 4: ARIMA prediction and wavelet power spectrum of USA Data.

Results

An analysis of tornado events in the United States between May 6, 1952 and September 10, 2025, reveals similar fluctuations in tornado frequency, with notable peaks around 1973 and 1998. A recurring cycle of approximately 24 to 28 years suggests nationwide tornado activity follows a predictable yet divergent pattern. Wavelet analysis confirms this periodicity. Auto ARIMA forecasting predicts a likely increase in tornado activity through 2027.

Discussion & Conclusion

Our analysis reveals a recurring 24–28 year cycle in tornado activity, significantly different from the observed 13–17 year cycle observed in New York State. As tornado activity varies greatly nationwide, it is possible that geographic subregions may yield more similar results. Future research will test this hypothesis.

ARIMA modelling suggests that the United States will, generally, experience more tornado events in the upcoming years. This is unlike the forecast given for New York State, which instead expects a decrease in tornado events. There also exists a sizable increase in tornado activity between 1952 and 1973, which is the likely result of better detection and reporting.

Tornadoes are difficult to predict, costing the United States billions of dollars in damages, directly claiming the lives of 51 people, and causing hundreds on injuries in 2024 alone. Further, they have multiple negative environmental effects, contributing to habitat destruction, soil erosion, and water and air pollution. The ability to predict years with heightened or lessened tornado activity will allow better counteraction of these consequences.

References

1. Hubbs, D. et al. (2025). A Preliminary Approach to Investigate Recent Increases in New York Tornado Activity: A Wavelet Analysis and Forecasting Approach using R.
2. Roesch, A., & Schmidbauer, H. (2018). WaveletComp: Computational Wavelet Analysis. R package version 1.1. <https://CRAN.R-project.org/package=WaveletComp>.
3. Chao, K. S., & Ripley, B. (2022). TSA: Time Series Analysis. R package version 1.3. <https://CRAN.R-project.org/package=TSA>.
4. Wickham, H., Hester, J., & Rye, J. (2024). readr: Read Rectangular Text Data. R package version 2.1.5. <https://CRAN.R-project.org/package=readr>.
5. Spira, V., Grolmund, G., & Wickham, H. (2023). lubridate: Make Dealing with Dates a Little Easier. R package version 1.9.3. <https://CRAN.R-project.org/package=lubridate>.
6. Wickham, H., François, R., Henry, L., & Müller, K. (2023). dplyr: A Grammar of Data Manipulation. R package version 1.1.4. <https://CRAN.R-project.org/package=dplyr>.
7. Zhou, H., & Wong, W. K. (2021). kza: Kolmogorov-Zurbenko Adaptive Filters. R package version 4.2. <https://CRAN.R-project.org/package=kza>.
8. Zeileis, A., & Grothendieck, G. (2005). zoo: S3 Infrastructure for Regular and Irregular Time Series. Journal of Statistical Software, 14(6), 1–27. <https://doi.org/10.18637/jss.v014.i06>.
9. Hyndman, R., & Khandakar, Y. (2008). Automatic Time Series Forecasting: The forecast Package for R. Journal of Statistical Software, 22(3), 1–22. <https://doi.org/10.18637/jss.v022.i03>.
10. Chang, W., & Park, W. (2024). scorepeak: Scoring and Ranking Peaks of Curves. R package version 1.1.0. <https://CRAN.R-project.org/package=scorepeak>.
11. NCEP. "Storm Events Database | National Centers for Environmental Information." NOAA.gov, 2025. <https://www.ncep.noaa.gov/stormevents/>.
12. Gilbert, Nathan. "Environment Pollution and Climate Change - Tornadoes and Their Impact on the Environment: Unleashing Nature's Fury." Environment Pollution and Climate Change, 29 July 2023. <https://www.environmentpollutionandclimatechange.com/tornadoes-and-their-impact-on-the-environment-unleashing-natures-fury-125477.html>.
13. National Oceanic and Atmospheric Administration. (2021, March 18). El Niño, La Niña, and tornadoes. NOAA Climate.gov. <https://www.climate.gov/news-features/spotlights/tornadoes>.
14. Liu, Shun, Ming Xue, and Qin Xu. "Using Wavelet Analysis to Detect Tornadoes from Doppler Radar Radial-Velocity Observations." Journal of Atmospheric and Oceanic Technology 24.3 (2007): 344–359. <https://doi.org/10.1175/JCLI1319.1>.
15. Lu, X. et al. (2018). Changes in the Seasonality of Tornado and Favorable Genesis Conditions in the Central United States: Changes of Tornado Seasonality & Genesis. Geophysical Research Letters, 45(10). <https://doi.org/10.1029/2018GL079598>.

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